

ONIQ Step 11D — ARAP Failure Diagnosis + Still-Key Investigation

Date: 2026-09-02
Measurement run: 33658060198
WALKING QC run: 33658066741
Artifact: 9857703837

Decision preserved: ARAP_AS_SELECTIVE_PROVIDER . No threshold change, no provider enablement, no production code changes.

Verified real-gateway result

Metric	Value
Supplied records	21
Measured-valid records	9
Eligible	2 / 9 = 22.2%
Rejected	7 / 9 = 77.8%
Wilson 95% CI	6.32% - 54.74%
Eligible WALKING QC	0 / 2 passed

A. Motion failure analysis

Shot 002 — 87c2b756-a832-48dc-b97b-7a3235ddee5c-s-shot002

Verdict: INTERACTION

The character translates across the canvas during the WALKING cycle, contacts the right edge, and finally vanishes.

Quantity	Observation
Frames	269 body frames (270 total)
Initial bbox	[158, 43, 354, 392]
Final bbox	[499, 301, 499, 317] (width collapsed to 1 px)

Centroid x range	256.0 → 499.0 px (range 243 px)
Mean centroid x velocity	+0.91 px/frame
Foot line range	317 – 397 px (80 px)
Edge contact	Frames 159 – 178 (right edge)
Vanished	Frames 237 – 256
Mean fill	12.53%

Out-of-mesh rig points (shot 002)

Normalized point	Nearest joint	Distance
(0.584, 0.045)	neck	0.128
(0.859, 0.267)	left_shoulder	0.024
(0.872, 0.176)	left_hand	0.057
(0.896, 0.072)	left_hand	0.133

The left_shoulder point is the closest match. These joints are in the upper-left limb and lie outside the silhouette mesh; they drive no triangles. They are evidence of mesh coverage, not the sole cause of the frame exit.

Shot 004 — 87c2b756-a832-48dc-b97b-7a3235ddee5c-s-shot004

Verdict: INTERACTION

Same global drift pattern, starting further right and with lower fill.

Quantity	Observation
Frames	589 body frames (590 total)
Initial bbox	[193, 141, 315, 362]
Final bbox	[499, 303, 499, 303] (width collapsed to 1 px)
Centroid x range	253.5 → 499.0 px (range 245.5 px)
Mean centroid x velocity	+0.42 px/frame
Foot line range	303 – 366 px (63 px)
Edge contact	Frames 346 – 365 (right edge)
Vanished	Frames 526 – 545
Mean fill	4.88%

Architectural conclusion

The WALKING renderer produces locomotion that is not frame-anchored. The character’s centroid drifts right, the foot baseline wanders, and the silhouette eventually exits the canvas. The isolated QC gate correctly flags this as “left frame / unstable feet.”

Do not loosen the 65% eligibility gate. The failure is not that too few characters are admitted; it is that the admitted characters do not stay inside the frame under the current WALKING implementation.

Follow-up proposals (documented only)

1. Centering correction: keep the character centroid fixed in the canvas each frame.
2. Foot-ground stabilization: constrain the lowest silhouette row to a narrow band.
3. Mesh coverage: ensure shoulder/hand joints land inside the triangulated silhouette, or filter unused out-of-mask joints.

B. Missing still-key investigation

The nine missing keys belong to the newest film:

`jobId = 64874747-44af-4086-a58a-250d2e7eec08` , scene `s` , shots `shot000 ... shot008` .

Key lifecycle findings

Stage	Finding
Key derivation	Correct. <code>stillIdFor(jobId, sceneId, shotId)</code> and <code>stillKeyFor</code> produce <code>story/still/<jobId>-s-shotNNN.png</code> . Verified in source and tests.
Worker request	Correct. <code>story-worker.mjs</code> passes the same identifiers to <code>drawStill</code> and later to <code>story-motion</code> .
story-still edge function	Correct. Derives <code>stillId</code> from verified job token + body scene/shot.
Gateway storage	Correct now. <code>keepStill</code> writes to <code>story/still/<id>.png</code> and verifies with HEAD.
R2 object presence	NOT_OBSERVABLE from this sandbox. Public probe returned HTTP 000 (network unreachable).
Measurement fetch	HTTP 404 for all nine <code>64874747</code> stems on run 33646171242.

Classification	MISSING — listed in manifest, absent from driver output.
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Root cause

The manifest provenance note records the cause:

“Run 33646171242 (2026-09-02): 9 of 18 keys answered 200 — all nine shots of the 2026-08-31 film — and all nine of the 2026-09-02 film answered 404.”

Film 64874747 rendered before the gateway still path was fixed to write the derived key. The fix shipped afterwards; the historical frames were not retroactively uploaded.

Divergence point for each missing key

Expected key	First divergence
64874747-...-s-shot000.png	Upload / persistence
64874747-...-s-shot001.png	Upload / persistence
64874747-...-s-shot002.png	Upload / persistence
64874747-...-s-shot003.png	Upload / persistence
64874747-...-s-shot004.png	Upload / persistence
64874747-...-s-shot005.png	Upload / persistence
64874747-...-s-shot006.png	Upload / persistence
64874747-...-s-shot007.png	Upload / persistence
64874747-...-s-shot008.png	Upload / persistence

Follow-up proposals (documented only)

1. Leave the keys classified MISSING . They are excluded from the eligibility denominator correctly.
2. If the owner authorizes it, backfill the original frames under the derived keys to complete the corpus.

Final state

- Decision remains ARAP_AS_SELECTIVE_PROVIDER .
- allowPoseWarp: false preserved.

- `maxBboxFillPct: 65` preserved.
- No thresholds loosened, no grammar expanded, no GPU infrastructure introduced.
- No production code changes made in this step.

Generated by GitHub Copilot — ONIQ Step 11D diagnostic report.